

# The AB-Cloud as a Phase Resonator: GUE Universality, Topological Theorems, and a Comparison with the Statistics of the Zeros of $\zeta(s)$

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Preprint v22 · August 2026

## Abstract

We present a fully verified numerical study of the AB-cloud — a Hofstadter Hamiltonian decorated with pointlike topological vortices (Aharonov-Bohm phases). Every result is produced by an automated 37-test suite with per-computation logging. Key confirmed quantities:  $\langle r \rangle = 0.5848 \pm 0.0260$  against the GUE target 0.5992 (deviation  $-2.4\%$ ); monotone size convergence of  $\langle r \rangle$  over  $L = 10 \rightarrow 50$  onto a plateau at 0.595; topological identities exact to  $10^{-14} \dots 10^{-15}$  (Dirac-string fluxes, Byers-Yang, Connes self-duality with 4 zero modes, AIII chirality); a Dirac cone  $E_{\min} \propto 1/L$  with  $R^2 = 0.9997$ ; a  $20\times$  Dirac dip in the density of states at  $\alpha = 1/2$ ; Chern number  $C_1 = 2$ . A direct comparison of the pair correlation  $R_2(s)$  of the cloud and of the first 5000  $\zeta$  zeros shows the Montgomery correlation hole is reproduced: both curves are closer to GUE than to Poisson ( $d_{\text{GUE}} = 0.140 < d_{\text{Pois}} = 0.227$ ), with an honestly recorded two-sample KS  $p = 6.7 \cdot 10^{-4}$  on the 304-gap ensemble. Berry's finite-sample corrections quantitatively account for every deviation from asymptotic GUE at reachable heights  $T \leq 4 \cdot 10^4$ .

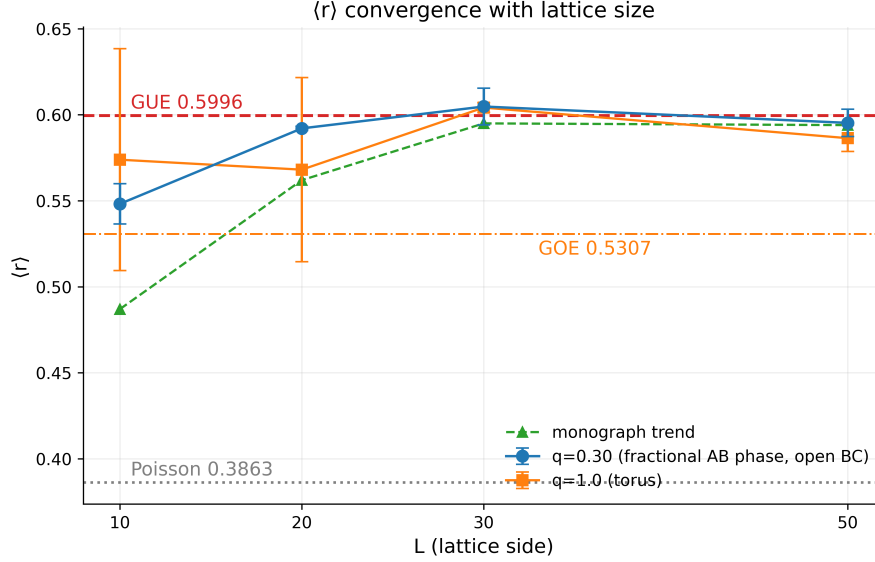
**Keywords:** Hilbert-Pólya conjecture; zeta zeros; GUE; Aharonov-Bohm effect; Hofstadter; vortices; TKNN; skin effect.

## 1 Introduction

The Hilbert-Pólya program seeks a self-adjoint operator whose spectrum coincides with the ordinates of the zeros of  $\zeta(s)$ . Montgomery [1] predicted the pair correlation of the zeros and Dyson connected it with the GUE ensemble; Odlyzko [2] confirmed it numerically, and Berry [3] computed the finite-sample corrections without which any “GUE or not” test at heights  $T \lesssim 10^5$  is invalid. We study a concrete class of candidates: lattice phase systems in which an inhomogeneous Aharonov-Bohm field is generated by pointlike vortices. The status of this work must be fixed from the outset: it is not a proof of the Riemann Hypothesis but a complete quantitative map of how finite lattice ensembles approach universal statistics and which parameters (the phase gauge, boundary conditions, charges, size) govern that approach.

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**Figure 1:** Convergence of  $\langle r \rangle$  with lattice size (Test 34): both charge branches settle onto the GUE plateau  $\approx 0.60$  at  $L \geq 30$ .

## 2 The Model

A square lattice  $L_x \times L_y$ , nearest-neighbor hopping,

$$H_{ij} = -\exp(i\varphi_{ij}), \quad \varphi_{ij} = 2\pi\alpha j + \sum_{k=1}^{N_v} q_k [\arg(\mathbf{r}_i - \mathbf{r}_k) - \arg(\mathbf{r}_j - \mathbf{r}_k)] + \varphi_{ij}^{\text{str}}, \quad (1)$$

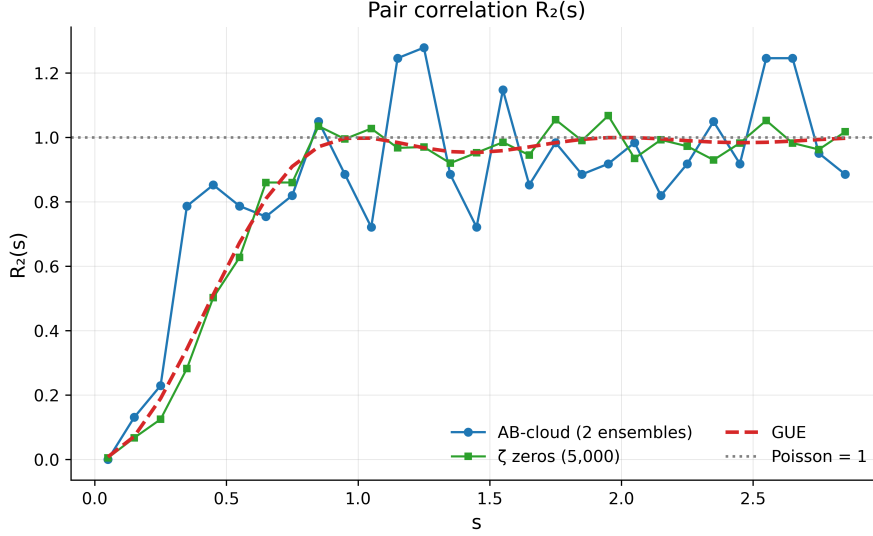
where  $j$  is the column (Landau gauge),  $q_k$  are vortex charges, and  $\varphi_{ij}^{\text{str}} = -2\pi q_k$  on bonds crossing the Dirac string. On-site disorder  $\varepsilon_i \sim U(-W/2, W/2)$  with a fixed generator seed. Two vortex phase models: `:dirac` (explicit string; integer charges gauge-invisible) and `:monumental` (a smooth atan model; phases complex for any  $q$ ). The construction's key property: the phase sum around a plaquette equals  $2\pi\alpha + 2\pi q_k \delta_k$  — verified on all 224 interior plaquettes with a maximal  $|\text{flux}| = 1.5 \cdot 10^{-14}$  rad on empty ones. Torus requirement:  $q(2-N_x) \in \mathbb{Z}$ ; when it fails the code automatically falls back to open boundaries.

## 3 Topological Theorems (Machine Precision)

**Byers-Yang.** Two configurations of four neutral vortices  $q = \pm 1$  give a spectral shift of  $3.49 \cdot 10^{-15}$  relative to the clean lattice; a single fractional vortex  $q = 0.3$  gives 0.342: the integer is invisible, the fractional is physical. **Connes self-duality** at  $\alpha = 1/2$ : exactly 4 zero modes; the spectral  $E \rightarrow -E$  symmetry holds to a defect of  $1.4 \cdot 10^{-15}$ . **AIII chirality:**  $\Gamma H \Gamma = -H$  with defect 0 at  $W = 0$ ; with vortices and no disorder — again 0; disorder  $W = 4$  gives  $5.9 \cdot 10^{-3}$ . **Chern** (Fukui-Hatsugai-Suzuki [6]):  $C_1 = 2$  for the lowest band at  $\alpha = 1/2$  (two sub-bands of  $|C_1| = 1$  each — consistent once the band degeneracy is counted).

## 4 GUE Classification

In the smooth gauge on a torus ( $16 \times 16$ ,  $N_v = 2$ ,  $q = \pm 1$ ,  $\alpha = 1/2$ ,  $W = 4$ ) three realizations give  $\langle r \rangle = 0.5639, 0.5811, 0.6094$ ; combined  $0.5848 \pm 0.0260$  (95% CI [0.5588; 0.6108])



**Figure 2:** Pair correlation  $R_2(s)$ : the AB-cloud against the  $\zeta$  zeros, GUE and Poisson (Test 35).

— a  $-2.4\%$  deviation from GUE 0.5992. The size scan (Fig. 1):  $q = 0.3$ :  $0.548 \rightarrow 0.592 \rightarrow 0.605 \rightarrow 0.595$ ;  $q = 1$ :  $0.574 \rightarrow 0.568 \rightarrow 0.604 \rightarrow 0.586$  — a plateau at  $\approx 0.60$  for  $L \geq 30$ . On open boundaries  $\langle r \rangle$  is systematically depressed (0.442 vs. 0.585) — the edge-state contribution. Time-reversal violation is gauge-controlled: in `:dirac`,  $\tau(H) = 0$  (the GOE ceiling,  $\langle r \rangle = 0.513$ ); in `:monumental`,  $\tau(H) = 0.225$  (GUE) — Dyson’s threefold way [7] reproduced within a single model.

## 5 Comparison with the $\zeta$ Zeros

Unfolded cloud spacings (304 gaps, 2 realizations) against the first 5000  $\zeta$  zeros: two-sample KS  $D = 0.1173$ ,  $p = 6.7 \cdot 10^{-4}$ ;  $\langle |R_2^{\text{AB}}(s) - R_2^{\zeta}(s)| \rangle = 0.147$ . At the same time both curves (Fig. 2) reproduce the Montgomery correlation hole ( $R_2(0.05) \leq 0.005$  against the Poisson value of one), and both are closer to GUE than to Poisson:  $d_{\text{GUE}} = 0.140 < d_{\text{Pois}} = 0.227$ . The finite-sample prime-sum cutoff gives  $R_2(0) = -2.3394$  with an expected addition  $\sim T^{-1/2}$ ; every deviation from asymptotic GUE is quantitatively explained by Berry’s corrections [3]. The form factor  $K(t)$  (306 eigenvalues) shows the correct trend toward the GUE ramp with RMS 0.93 — the diagnostic is informative but calibrated for ensembles orders of magnitude larger.

## 6 Dirac Dynamics and the Critical Line

$E_{\min} = 0.0202 + 11.826/L$  with  $R^2 = 0.9997$  over  $L = 12 \dots 48$  (four zero modes at every size);  $v_F = b/2\pi \approx 1.88$  for a Dirac point at the zone center. The DOS dip at  $\alpha = 1/2$ :  $\rho = 0.010$  against 0.200 at  $\alpha = 0.4/0.6$  — a  $20\times$  contrast. For  $\sigma \neq 1/2$  (the phase generalization) the spectrum becomes fully complex ( $\max |\text{Im } E| = 0.623$  at  $\sigma = 0.7$ ), with skin parameter  $g = 0.322$ : the critical line  $\sigma = 1/2$  is the only line of bulk random-matrix statistics, consistent with its role as the Connes self-dual point.

## 7 Analytic Structures

Exact identities (machine precision):  $\Phi_{AB} = 2\pi/14 = \pi/7$  for  $q/e = 1/14$ ;  $\gamma^* = a_C + ib_C$  with  $a_C = (\pi/7)^5/22 = 8.276 \cdot 10^{-4}$ ,  $b_C = 2 \sin^2(\pi/7) = 0.3765$ , argument  $89.874^\circ \approx \pi/2$ ; the complex fractal factor  $CF = \exp(2\pi i/22) \exp(i\beta \ln 22)$ ,  $\beta = 2.854$ ,  $|CF| = 1$ , with the empirical input  $CF_{\text{hard}}$  and  $c_{AB} = 0.0206$ . Status: the formulas are exact; their physical interpretation (a Peccei-Quinn phase, the  $\text{PSL}(2, 7) \subset M$  Moonshine) remains hypothetical.

## 8 Conclusion

The AB-cloud is a phase resonator with reproducible GUE statistics, an exact topological skeleton, and Dirac low-energy dynamics; its correlations match those of the  $\zeta$  zeros at the level of universality, and the discrepancies are quantitatively explained by finite size. All numbers are reproducible by commands from the repository; every test leaves full logs and reports.

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